

# Stable Circular Orbits in Anti-de Sitter Schwarzschild Spacetime

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## Abstract

The two important aspects of stable circular motion, namely the angular frequencies of orbital and small radial oscillations, in Anti de Sitter–Schwarzschild Spacetime are discussed and analyzed. We use the Effective Potential concept in our analysis and investigate the conditions for forming of stable circular orbits for uncharged massless (photon) and material (massive) test particles. As we will see the conditions impose constraints on current parameters.

*Keyword:* General Relativity; Anti de Sitter Spaces; Schwarzschild Spacetime; Small Radial Oscillations Frequency; Orbital Frequency; Effective Potential.

## 1 Introduction

The Einstein's Field Equations with Cosmological Constant (CC)  $\Lambda$  is:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}, \quad (1)$$

where  $R_{\mu\nu}$ ,  $R$ ,  $g_{\mu\nu}$  and  $T_{\mu\nu}$  are the Ricci Tensor, Ricci Scalar, metric Tensor and the Energy–Momentum Tensor, respectively. In the context of modern cosmology, for the universe to be accelerating, it is necessary for CC to be positive which plays as a repulsive form of gravity. The CC is the simplest possible explanation for dark energy (which constitutes approximately sixty–eight percent of the mass–energy density of the universe) in the standard model of cosmology, model known as the  $\Lambda$ CDM. In the modern approach,  $\Lambda$  is considered as a form of energy present even in empty space (known as Vacuum Energy), arising from an expanding model known as de–Sitter universe,<sup>1</sup> a universe without radiation and matter ( $T_{\mu\nu} = 0$ ), thought to correspond to dark energy in the inflationary (early) universe.

On the other hand, for spherically symmetric spacetime, the Vacuum Solution (to the Einstein's Field Equations) was first provided by Karl Schwarzschild in late 1915. This solution is for the vacuum region of spacetime outside a single central massive body, given by [1]:

$$ds^2 = \left(1 - \frac{r_s}{r}\right) dt^2 - \frac{dr^2}{1 - \frac{r_s}{r}} - r^2 \left(d\theta^2 + \sin^2\theta d\phi^2\right), \quad (2)$$

where  $r_s = 2GM/c^2$  is the Schwarzschild radius<sup>2</sup>

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<sup>1</sup>The scale factor for the flat de–Sitter universe is proportion to:  $\exp\left(\sqrt{\Lambda/3}t\right)$ .

<sup>2</sup>In the Schwarzschild radius,  $M$  is the mass of the central body,  $G$  and  $c$  are the gravitational constant and speed of light, respectively. Throughout this article we will use geometric units in which  $G = c = 1$  and therefore  $r_s = 2M$ .

Considering the Vacuum Energy ( $\Lambda \neq 0$ ) in this case, the resulting solution is known as de Sitter–Schwarzschild metric given by:

$$ds^2 = \left(1 - \frac{r_s}{r} - \frac{\Lambda r^2}{3}\right) dt^2 - \frac{dr^2}{\left(1 - \frac{r_s}{r} - \frac{\Lambda r^2}{3}\right)} - r^2 \left(d\theta^2 + \sin^2 \theta d\phi^2\right). \quad (3)$$

The de Sitter–Schwarzschild metric with  $\Lambda < 0$  is known as Anti de Sitter–Schwarzschild metric. The Physical and Mathematical aspects of the spacetime metric described by (3) are studied in many of works for both (positive and negative) values of  $\Lambda$ , e. g. [2, 3, 4, 5]. In this work, we are interested in the Anti deSitter (**AdS**) case ( $\Lambda < 0$ ), and this is because, on the one hand, in mathematics and physics, AdS space is a maximally symmetric Lorentzian manifold with constant negative scalar curvature and the other hand, one of the most important features of the asymptotically (AdS) spacetime is the preservation of the AdS symmetry structure at infinity. By incorporating this asymptotic symmetry into the corresponding Hamiltonian, it is possible to study gravitational dynamics in the presence of negative energy. It should be noted, the asymptotic symmetry doesn't appear for ( $\Lambda > 0$ ) [6].

The asymptotically AdS spacetimes play a central role in theoretical physics which is (primarily) due to the AdS/CFT correspondence (Theory)<sup>3</sup> There is a long list of work done on asymptotically AdS spacetimes of which we will only mention a few here, e. g. [7, 8], or specially for Regular asymptotically AdS Black holes[9, 10].

In the present work, our goal is to investigate the gravitational dynamics of an important aspect of circular motion, namely the frequency of small radial oscillations, and also its relationship with the orbital frequency is examined.

Circular motion are the simplest form of orbital motion in celestial mechanics, characterized by a rotating body maintaining a constant radius while orbiting a central gravitational mass. It refers to the motion of celestial bodies, such as planets, moons, satellites, along circular or nearly circular orbits under the influence of gravity. Most of the stars in the galactic disk move in roughly circular orbits around the galactic centre. These orbits are also especially significant in the equatorial plane, such as in the structure of accretion disks around astrophysical black holes[11].

One of the most important features of circular orbits is its stability which is discussed and analyzed in this paper for massive and massless (photon) particles in AdS–Schwarzschild spacetime. The results show that the *AdS case is more compatible with physical universe, than the de Sitter case*. This consistency holds more generally for asymptotically AdS spaces as well, as shown for the thermodynamic properties[12].

One of the most capable background for analyzing circular motion is spherically symmetric spacetime, because in such a case, through a concept called Effective Potential (**EP**), the motion can be analyzed with its one-dimensional counterpart. The EP method is a powerful tool for describing the motion of a free particle in the equatorial plane of a Schwarzschild–like center of attraction. The EP encapsulates many fundamental aspects of particle dynamics, serving as a classification scheme for orbit configurations. Also, the graph of the EP can be used to derive several interesting qualitative information about the orbits. For instance, stable (unstable) circular orbits in the equatorial plane correspond to the minimum (maximum) points of the EP.

In many practical applications in classical and relativistic mechanics, the gravitational field from a spherically symmetric source (central body) is relevant for many important physical situations, hence the EP method is one of the most useful tools in these situations. Below is a simple summary of the use of the EP method in this article.

In the static spherically symmetric case the spacetime line element can be written as:

$$ds^2 = f(r) dt^2 - \frac{dr^2}{f(r)} - r^2 \left(d\theta^2 + \sin^2 \theta d\phi^2\right), \quad (4)$$

<sup>3</sup>The AdS/CFT correspondence is a conjectured relationship between two types of physical theories: AdS spacetime and invariant Conformal Field Theory (CFT) living on the boundary of this spacetime. It was first proposed by theoretical physicist Juan Maldacena in 1997 and has since become a central idea in string theory and theoretical physics.

where the metric function  $f(r)$  is determined by the associated field equations. It is shown that, the EP corresponding to the metric function  $f(r)$  becomes [13]:

$$V(r) = \frac{1}{2} \left( \epsilon + \frac{L^2}{r^2} \right) f(r), \quad (5)$$

where  $\epsilon = 0, 1$  corresponds to the photon (massless) and material (massive) particles, respectively and  $L$  is the angular momentum constant.

The Small Radial Oscillations is an important forms of radial motion in classical mechanics, a useful concept to understand how objects move when they are only slightly perturbed from a circular orbit. The object oscillates inward and outward in radial direction, while it revolves around the central body. It is well known that the small radial oscillations occur for stable circular orbits which are possible if the following relationships hold:

$$V'(r_{min}) = 0 \quad \text{and} \quad V''(r_{min}) > 0,$$

where the prime denotes derivative of EP with respect to the radial coordinate  $r$  and  $r_{min}$  is the radius of the stable circular orbit (minimum EP point).

For a more complete analysis, the orbital frequency is also discussed in parallel, because it is a remarkable feature of circular orbits. It is one of the fundamental quantities in the motion such that it can be used to analyze or obtain many properties of motion or objects involved in motion. To determine the orbital frequency, we know that for motion in spherically symmetric potential, the angular constant of motion  $L$ , corresponding to the  $\phi$  coordinate is given by[1]:

$$\dot{\phi}(r) = \frac{L}{r^2}, \quad (6)$$

which for circular motion gives the orbital frequency as:

$$\omega = \dot{\phi}(r_{min}) = \frac{L}{r_{min}^2}, \quad (7)$$

On the other hand, by the Taylor expansion of the EP around the  $r_{min}$ , it is shown that the square of small radial oscillations frequency  $\omega_r^2$  is proportion to second derivative of the EP at  $r_{min}$ , that is[14]

$$\omega_r^2 = V''(r_{min}), \quad (8)$$

where the proportionality coefficient is chosen to be equal to unity. Formulas (7) and (8) are basic for subsequent calculations in this work.

The work is organized as follows:

In section 2, the stability of circular orbit, the radial oscillations and orbital frequencies in AdS-Schwarzschild geometry are discussed and analyzed. The conclusions are given in the last section.

## 2 Stable Circular Orbit

To begin the discussion, let us calculate the EP function for the the AdS-Schwarzschild metric function, that is  $f(r) = 1 - \frac{r_s}{r} - \frac{\Lambda r^2}{3}$ , where  $r_s = 2M$ . By substituting the metric function into Formula (5), the EP becomes:

$$V(r) = \frac{1}{2} \left( \epsilon + \frac{L^2}{r^2} \right) \left( 1 - \frac{r_s}{r} - br^2 \right), \quad (9)$$

where  $b = \Lambda/3$ .

For  $\epsilon = 0$ , the last equation is:

$$V(r) = \frac{L^2}{2} \left( \frac{1}{r^2} - \frac{r_s}{r^3} - b \right), \quad (10)$$

whose first and second derivatives are:

$$V'(r) = \frac{L^2}{2r^3} \left( \frac{3r_s}{r} - 2 \right), \quad (11)$$

$$V''(r) = \frac{3L^2}{r^4} \left( 1 - \frac{2r_s}{r} \right), \quad (12)$$

By solving equation  $V'(r) = 0$ , we find  $r = r_m = (3/2)r_s$ , where  $r_m$  represents the extremum point. Since  $V''(r_m) < 0$ , then  $r_m$  is the maximum point corresponding to an unstable orbit. Therefore, the result is that in both (AdS and de Sitter) cases, there is no stable orbit for the massless particle (photon). It is easily to verify that this result also holds in the Schwarzschild geometry, that is  $b = 0$ . For  $\epsilon = 1$ , EP (9) becomes:

$$V(r) = \frac{1}{2} \left( -\frac{L^2 r_s}{r^3} + \frac{L^2}{r^2} - \frac{r_s}{r} - br^2 + 1 - bL^2 \right), \quad (13)$$

which the first and second derivatives are:

$$V'(r) = \frac{1}{2} \left( \frac{3L^2 r_s}{r^4} - 2\frac{L^2}{r^3} + \frac{r_s}{r^2} - 2br \right), \quad (14)$$

$$V''(r) = \frac{-6L^2 r_s}{r^5} + \frac{3L^2}{r^4} - \frac{r_s}{r^3} - b. \quad (15)$$

As previous, to find the extremum points of the EP, we must solve the equation  $V'(r) = 0$ , which after simplification leads to the following equation:

$$-2br^5 + r_s r^2 - 2L^2 r + 3L^2 r_s = 0. \quad (16)$$

This equation is a fifth-degree equation that in general may be unsolvable or have complicated solutions that are not useful for our purposes. However, since the solutions to this equation determine the minimum and maximum points of the EP and *we are only looking for a minimum solution*, to avoid the aforementioned complications, we achieve our goal of finding the minimum solution by making some assumptions on the coefficients of the equation, as shown below<sup>4</sup>. The assumptions are made for both (positive and negative) values of  $\Lambda$ . The partial review of case  $\Lambda > 0$  is order to show that, it is the case  $\Lambda < 0$  that leads to a greater number of suitable solutions. It should be noted that the assumptions used are based on the form of the fifth-degree equation in both cases and using an algebraic theorem which states: every odd-degree equation has at least one real solution.

#### Positive $\Lambda$ : $b > 0$

The algebraic theorem mentioned above can also be stated as follows:

Every fifth degree expression can always be written as the product of a first degree expression and a fourth degree expression, that is:

$$\text{fifth degree expression in } r = (r - r_1)(\text{fourth degree expression in } r). \quad (17)$$

By this theorem, to find (at least) a one root of equation (16), we can write it as follows:

$$2br^2(r^3 - \frac{r_s}{2b}) + 2L^2(r - \frac{3r_s}{2}) = 0. \quad (18)$$

Now by assuming  $\frac{3r_s}{2} = r_1$  and  $\frac{r_s}{2b} = r_1^3$  (which implies  $9\Lambda M^2 = 1$ ), equation (18) reads:

$$br^2(r^3 - r_1^3) + L^2(r - r_1) = (r - r_1)(\text{fourth degree expression}) = 0, \quad (19)$$

<sup>4</sup>The coefficient of equation (16) are in terms of important physical parameters, that is  $M, L, \Lambda$ .

where by using Algebraic Identity ( $r^3 - r_1^3 = (r - r_1)(r^2 + r_0r + r_1^2)$ ), and Algebraic Theorem (17), the *fourth degree expression* becomes:

$$\text{fourth degree equation} = r^4 + r_1r^3 + r_1^2r + L^2/b. \quad (20)$$

Finally equation (18), can be written in the following form:

$$(r - r_1)(r^4 + r_1r^3 + r_1^2r + L^2/b) = 0, \quad (21)$$

but, the condition  $b > 0$ , dictates  $r^4 + r_1r^3 + r_1^2r + L^2/b > 0$ , indicating one and only one real root ( $r = r_1$ ) for equation (21) or the same (16). It is not hard to show  $V''(r_1) < 0$  by (15), meaning the maximum point corresponding to an unstable orbit. Therefore, the result is that under above assumption ( $r_1 = \frac{3r_s}{2} = \left(\frac{r_s}{2b}\right)^{1/3}$ ), there is no stable circular orbit for a test material particle, in de-Sitter Schwarzschild spacetime.

We would like to remind you that the last result obtained under the above assumption is not a general result, meaning that under other assumptions, equation (16) may have a minimum solution. However, since our discussion is devoted to AdS space, we limit ourselves to this brief discussion.

**Negative  $\Lambda$ :  $b < 0$**

In this case, using the following notation

$$b = \frac{\Lambda}{3} = -a^2, \quad l^4 = \frac{L^2}{a^2}, \quad (22)$$

where  $a$  is a positive number, we can write the equation (16) as follows:

$$2a^2r(r^4 - l^4) + r_s(r^2 + 3L^2) = 2a^2r(r^2 - l^2)(r^2 + l^2) + r_s(r^2 + 3L^2) = 0. \quad (23)$$

Now by assuming the

$$l^2 = 3L^2, \quad (24)$$

and inserting it into the equation (23) and dividing both sides by common factor  $(r^2 + l^2)$ , we get:

$$2a^2r(r^2 - l^2) + r_s = 0,$$

or

$$r^3 - l^2r + \frac{r_s}{2a^2} = 0. \quad (25)$$

Before discussing the roots of this cubic equation, note that from equations (22) and (24), we arrive at the following constraint equation:

$$9a^2L^2 = -9bL^2 = 1, \quad \text{or} \quad L^2 = -\frac{1}{3\Lambda}, \quad (26)$$

which is a constraint relation on current parameters  $L, \Lambda$ .

To discuss the roots of equation (25), we use the following theorem in algebra which states for the cubic equation of the form:

$$x^3 + px + q = 0, \quad (27)$$

if the two following inequalities hold:

$$p < 0 \quad \text{and} \quad \Delta = (p/3)^3 + (q/2)^2 < 0 \quad (28)$$

then the cubic equation has three real roots as:

$$x_k = 2\sqrt{-\frac{p}{3}} \cos\left(\frac{\alpha - 2k\pi}{3}\right), \quad \text{where} \quad \cos \alpha = \frac{3q}{2p} \sqrt{\frac{-3}{p}}. \quad (29)$$

By comparing (25) and (27), we find:

$$p = -l^2 = -3L^2 < 0, \quad q = \frac{r_s}{2a^2} \quad (30)$$

which by substituting into (second) inequality in (28) ( $\Delta < 0$ ), the following constraint relation is also attained

$$\frac{9r_s}{2L} < 1 \quad \text{or} \quad M < \frac{L}{9}. \quad (31)$$

The constrain relations (26) and (31) express the conditions to have a minimum point of EP (13), or equivalently, to form a stable circular orbit. Obviously, relation (26) is a constraint on values of the CC and test particle's angular momentum and relation (31) is a constraint on values of the central mass and test particle's angular momentum.

If the solution (29) is applied to (25), we get

$$x_k = 2\sqrt{-\frac{p}{3}} \cos\left(\frac{\alpha - 2k\pi}{3}\right), \quad (32)$$

where ( $k = 0, 1, 2$ ) and

$$\cos \alpha = -\frac{9r_s}{4L} = -\frac{9M}{2L}. \quad (33)$$

By a simple calculation or from the EP graph (shown in Fig.1), it is clear that the nature of the three extrema points are:

$k = 0$  : corresponds to a positive minimum,

$k = 1$  : correspond to a positive maximum,

$k = 2$  : correspond to a negative minimum,

therefore, the only acceptable solution is:

$$x_0 = r_{min} = 2L \cos\left(\frac{\alpha}{3}\right) = 2L \cos\left(\frac{\arccos\left(-\frac{9M}{2L}\right)}{3}\right), \quad (34)$$

where  $r_{min}$  stands for minimum point of EP.

Now that we have found the (acceptable) minimum point, we calculate the orbital angular frequency by (7), as

$$\omega = \frac{L}{r_{min}^2}, \quad (35)$$

and also small radial oscillations frequency by (8) and (15), as:

$$\omega_r^2 = V''(r_{min}) = -\frac{12L^2M}{r_{min}^5} + \frac{3L^2}{r_{min}^4} - \frac{2M}{r_{min}^3} + \frac{1}{9L^2}, \quad (36)$$

where  $r_{min}$  is given by (34) and the last term is due to (26). Note that by eliminating the  $r_{min}$  between the equations (35) and (36), the small radial oscillation frequency can be written as a function of the orbital frequency.

Evidently, the angular frequencies (35) and (36) are functions of the  $M$  and  $L$  which are constrained by (31), thus representing the both angular frequencies variations in terms of each of these parameters requires some rework. However, this problem can be mitigated by choosing new variables and angular frequencies. If we define the new variable as  $x = 9M/2L$  and the new (rescaled) angular frequencies

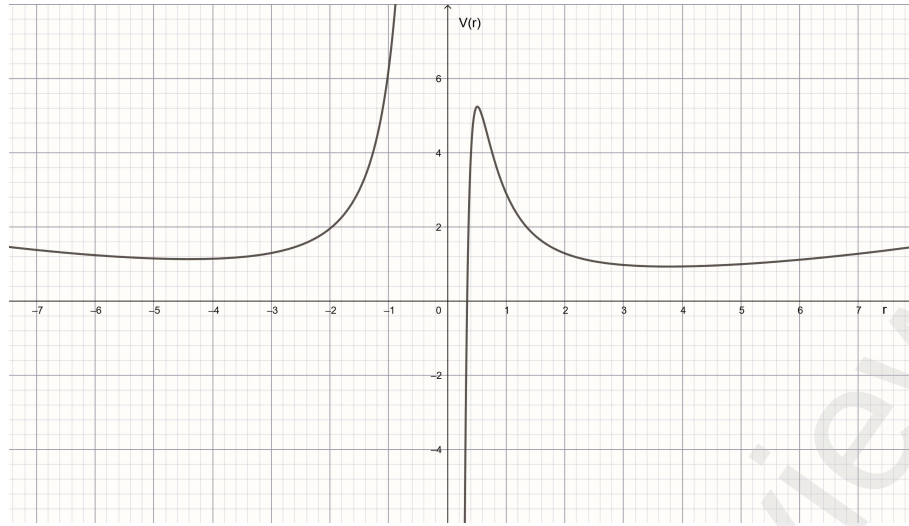


Figure 1: The graph of EP (13) versus  $r$ , for typical values  $L^2 = 8$  and  $r_s = 2M = 1/3$ . Also, note that in equation (14), due to (28), we put:  $-bL^2 = 1/9$ .

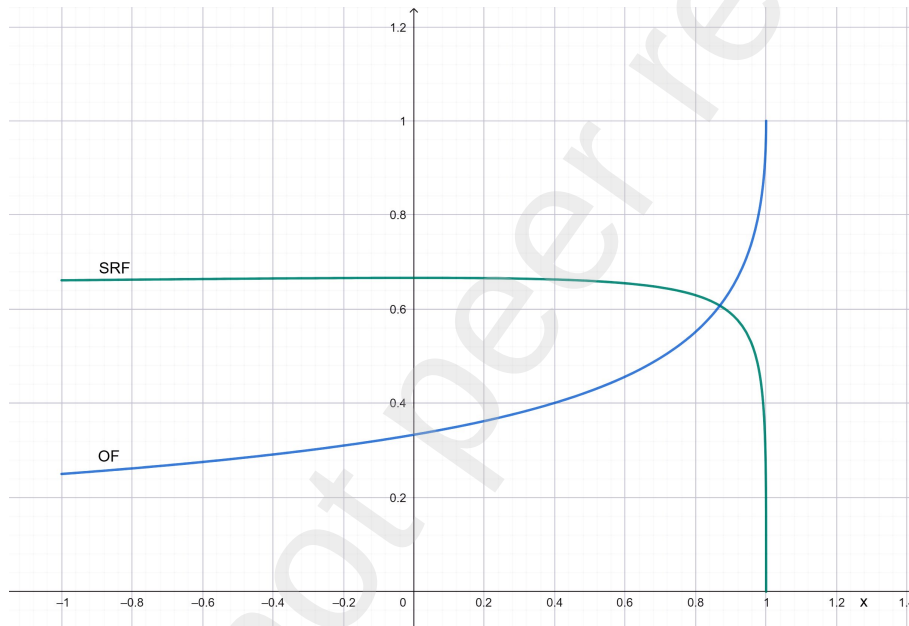


Figure 2: The graphs of the Orbital Frequency (OF)  $\Omega$ , and Small Radial Frequency (SRF)  $\Omega_r$ , given by (37) and (38), respectively.

by  $\Omega = L\omega$  and  $\Omega_r = L\omega_r$ , we can rewrite angular frequencies (35) and (36) as functions of  $x$  only, as follows:

$$\Omega(x) = \frac{1}{4y^2(x)}, \quad (37)$$

$$\Omega_r^2(x) = -\frac{x}{12y^5(x)} + \frac{3}{16y^4(x)} - \frac{x}{18y^3(x)} + \frac{1}{9}, \quad (38)$$

where  $y(x) = \cos\left(\frac{\arccos(-x)}{3}\right)$ .

The graphs of the functions (37) and (38), versus  $x$ , are illustrated in Fig.2. Looking at the figure, we can see the followings:

- 1) In the allowable Interval  $-1 < x < 1$ , the Orbital Frequency (OF) is increasing function and Small Radial Frequency (SRF) is a very slowly decreasing function, except at points close to  $x = 1$ .
- 2) The difference between two frequencies has a minimum at the beginning of the interval ( $x = -1$ ) and a maximum at the end of the interval ( $x = 1$ ).
- 3) The difference between two frequencies becomes zero at ( $x \approx 0.9$ ) which physically means that the duration of one complete radial oscillation is equal to the duration of one complete rotation of the particle.

Finally, we note that the new variable  $x = 9M/2L$  (introduced above), can be written in terms of CC, due to equation (26) as  $x = \frac{9\sqrt{3}}{2}M\sqrt{-\Lambda}$ . This form allows to show the variations of angular frequencies (37) and (38) versus CC.

### 3 Conclusions

In this work, the stable circular motion for an uncharged (massless and massive) test particle in AdS–Schwarzschild is considered, by focusing on two important features of circular motion, that is Small Radial Oscillations and Orbital Frequencies. The main conclusions can be briefly stated as follows:

- 1) As the Schwarzschild case ( $\Lambda = 0$ ), there is no stable orbit for massless (photon) particle.
- 2) The formation of a stable orbit (for material particle) imposes two constraints on the values of particle's angular momentum and central mass. The constraints are:  $\Lambda L^2 = -1/3$  and  $M < L/9$ .
- 3) In allowed Interval  $-1 < x < 1$ , where  $x = 9M/2L$ , the Orbital Frequency is an increasing function and Small Radial Frequency is a slowly decreasing function except at points close to  $x = 1$ , so that they become equal at a certain point, say  $x \approx 0.9$ . This physically means when particle's angular momentum is approximately five times of central mass ( $L \approx 5M$ ), the periods of one complete radial oscillation and one complete revolution become equal.

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